Chen 2020 NPC — analysis summary

RUN 2 OF 7

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Continuous (1 connected component) but NOT sample-mixed: ~50% of cells in patient-pure clusters at θ=2 (~90% at θ=1), robust across n_pcs {15,20,30} — not the attempt-1 n_pcs=6 artifact. Purest clusters = non-cycling P11 (patient-private CNV). Only θ=3 mixes, by over-integrating.

Gate 2 — module geography → NA

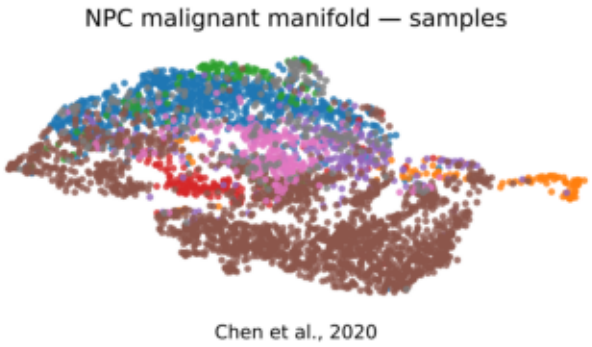
Not evaluated — no sample-integrated continuous manifold on which to fit a trajectory / score module geography.

Gate 3 — hub cross-repression → NA

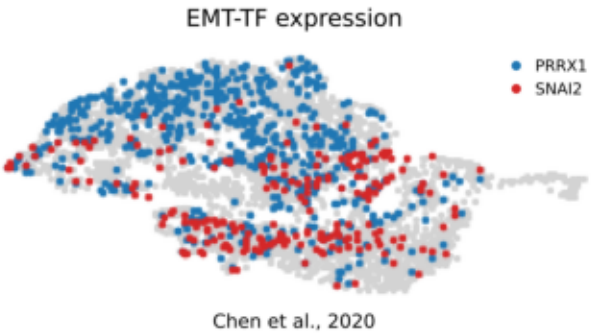
Not evaluated — requires a branching trajectory with an identified hub, which Gate 1 precludes.

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
Nasopharyngeal carcinoma (Chen et al., 2020), 10x — 14 patients / 71,896 cells (master scope); malignant only: 3,763 load → 3,674 across 8 patients (≥62 cells) enter the embedding.

Verdict.
Pattern does NOT reproduce — Gate 1 NO-GO. The malignant compartment is patient-segregated (inter-patient CNV heterogeneity), not a shared sample-integrated developmental manifold. No hub, no trajectory, no cross-repression axis to test. EMT-TFs shown for reference: PRRX1 / SNAI2 (FOXC/PRRX families; canonical SNAI1 & TWIST1 below the 3% detection floor).

Parameter choices

Min cells/sample	62 (median-restricted gap; keeps 8/14)	**Genes**	1,290 over-dispersed (GAM)
n_pcs	20 (re-derived; Kürten 20 / Puram 40)	**Integration**	Harmony on sample, θ=2, 20 it
Embedding	UMAP, Leiden res=0.5 → 9 clusters	**Trajectory**	none (Gate 1 NO-GO)
EMT-TF panel	detection-gated ≥3%; PRRX1/SNAI2/FOXC1	**Geometry**	patient-segregated (not a manifold)